

A disproof of Conjecture 00000005402

The Moebius conversion of periodic-orbit counts has no $\mu(d) \cdot d$ coefficients

1 The conjecture

Conjecture 00000005402 reads:

Definition: The Mobius conversion inverts the counts of all periodic orbits to recover the primitive (exactly periodic) orbit counts. *Conjecture:* The low-order conversion is always an explicit divisor-sum recursion with recursion coefficients small integer values of type $\mu(d)$ times d ; the transfer matrix of the recursion is triangular with diagonal entries 1; and the inverse conversion, the cumulative counting, is the partial sums of the primitive counts with nonnegative summation coefficients.

The Chinese text says the same; in particular the coefficients are of type $\mu(d)d$. The statement is a conjunction of three clauses:

- (i) (*coefficients*) at low order, the conversion from the total counts T_n to the primitive counts B_n is the divisor-sum recursion $B_n = \sum_{d|n} \mu(d) d T_{n/d}$;
- (ii) its matrix is triangular with diagonal 1;
- (iii) the inverse conversion has nonnegative coefficients.

We refute clause (i), already at order $n = 2$, which is the lowest order at which the clause says anything beyond $B_1 = T_1$. This makes the conjunction false. We do not need clauses (ii) and (iii), and we do not claim that they are false. The placement $\mu(d) d$ on $T_{n/d}$ is the one compatible with clause (ii): its diagonal coefficient ($d = 1$) is $\mu(1) \cdot 1 = 1$. Theorem 6 below also excludes the other placement at order 2.

2 Definitions and readings

A *finite dynamical system* is a map $f : \beta \rightarrow \beta$ on a finite type β . A point x is *periodic of period* n if $f^n(x) = x$. The *least period* $\text{mp}(x)$ is the smallest $n > 0$ with $f^n(x) = x$. Following Mathlib's `Function.minimalPeriod`, we set $\text{mp}(x) = 0$ when x is not periodic. The orbit of a periodic point x is the cycle $[x, f(x), \dots, f^{\text{mp}(x)-1}(x)]$ (Mathlib's `Function.periodicOrbit`). A *primitive orbit of length* n is the orbit of a point of least period n . The Moebius function satisfies $\mu(1) = 1$ and $\mu(2) = -1$; it is Mathlib's `ArithmeticFunction.moebius`.

The text does not say whether points or orbits are counted, so we use the following counts (Lean names in typewriter font), for $n \geq 1$:

- `perPts` f $n = \#\{x : f^n(x) = x\}$ (periodic points of period n);
- `exactPts` f $n = \#\{x : \text{mp}(x) = n\}$ (points of least period n);

- `primOrbits` f n = number of distinct orbits of the points with $\text{mp}(x) = n$;
- `divOrbits` f n = number of distinct orbits of the points with $f^n(x) = x$;
- `cumPts` f n = $\#\{x : 1 \leq \text{mp}(x) \leq n\}$;
- `cumOrbits` f n = number of distinct orbits of the points with $1 \leq \text{mp}(x) \leq n$.

Here `divOrbits` counts the periodic orbits whose length divides n . The last two are the “cumulative” counts read as ordinary partial sums. We test six readings of (total T , primitive B):

R1: (`perPts`, `primOrbits`) R2: (`perPts`, `exactPts`) R3: (`divOrbits`, `primOrbits`)
R1': (`cumPts`, `primOrbits`) R2': (`cumPts`, `exactPts`) R3': (`cumOrbits`, `primOrbits`).

For the readings R2 and R3 the classical Moebius inversion formula applies. It says that if $g(n) = \sum_{d|n} f(d)$ for all $n \geq 1$, then $f(n) = \sum_{d|n} \mu(d) g(n/d)$ [W1]. The resulting conversion has coefficients $\mu(d)$, not $\mu(d)d$. We use this only as motivation; the Lean proof below does not rely on it.

Definition 1 (`ConjecturedConversion T B N`). For every finite dynamical system f and every order $1 \leq n \leq N$, $B(f, n) = \sum_{d|n} \mu(d) d T(f, n/d)$ (the right side is `muDConv`).

3 Results

Lemma 2 (`muDConv_two`). $\sum_{d|2} \mu(d) d a(2/d) = a(2) - 2a(1)$.

Theorem 3 (`g_counterexample`). Let $g = (0)(1)(23)$ be the permutation of $\{0, 1, 2, 3\}$ with two fixed points and one 2-cycle. Then `perPts` = (2, 4), `divOrbits` = (2, 3) at $n = 1, 2$, and `exactPts` g 2 = 2, `primOrbits` g 2 = 1. The $\mu(d)d$ conversion predicts $4 - 2 \cdot 2 = 0$ (R1, R2) and $3 - 2 \cdot 2 = -1$ (R3), whereas the true values are 1, 2 and 1.

Proof. g is an involution. For an involution, $\text{mp}(x) = 1$ if $f(x) = x$ and $\text{mp}(x) = 2$ otherwise (`minimalPeriod_invol`, using Mathlib’s `minimalPeriod_eq_prime` with $p = 2$). The orbit is then $[x]$ or $[x, f(x)]$ (`periodicOrbit_invol`). After this rewriting every count is a finite decidable computation (`decide`). The orbits $[2, 3]$ and $[3, 2]$ are equal as cycles, so `primOrbits` g 2 = 1. $\square$

Lemma 4 (`perPts_eq_cumPts`, `divOrbits_eq_cumOrbits`). For $n \in \{1, 2\}$ and every finite system, $f^n(x) = x \iff 1 \leq \text{mp}(x) \leq n$. Hence `perPts` = `cumPts` and `divOrbits` = `cumOrbits` at orders 1 and 2.

Proof. $f^n(x) = x \iff \text{mp}(x) \mid n$ (Mathlib `isPeriodicPt_iff_minimalPeriod_dvd`). The divisors of 1 and of 2 are exactly the integers in $[1, 1]$ and $[1, 2]$. $\square$

Theorem 5 (`Main theorem`, `conjecture_5402_false`). For every $N \geq 2$, `ConjecturedConversion T B N` is false for each of the six readings R1, R2, R3, R1', R2', R3'.

Proof. Instantiate the claim at $\beta = \text{Fin } 4$, $f = g$, $n = 2$ and apply Theorem 3; for the primed readings, first rewrite the totals with the previous lemma. $\square$

The next result shows that the failure is not an accident of the witness: at order 2 the coefficients of any valid divisor-sum conversion are forced. We use the identity map of a one-point set and the swap of a two-point set.

Theorem 6 (`R1_no_integer_conversion`, `R2_coefficients_forced`, `R3_coefficients_forced`). Suppose that $B_2 = x T_2 + y T_1$ with $x, y \in \mathbb{Z}$ holds for every finite dynamical system.

- Under R1 there are no such integers: the one-point identity gives $0 = x + y$, and the swap gives $1 = 2x$.
- Under R2, $x = 1$ and $y = -1$ (from $0 = x + y$ and $2 = 2x$).
- Under R3, $x = 1$ and $y = -1$ (from $0 = x + y$ and $1 = x$).

Corollary 7 (`no_placement_of_muD`). If $\{x, y\} = \{\mu(1) \cdot 1, \mu(2) \cdot 2\} = \{1, -2\}$ (either placement of the order-2 coefficients), then $B_2 = xT_2 + yT_1$ fails for some finite system under each of R1, R2 and R3.

Proof. Under R1 this follows from the first item of Theorem 6. Under R2 and R3 we would need $y = -1$, but $-1 \notin \{1, -2\}$. $\square$

Remark 8. Under R2 and R3 the correct order-2 coefficient is $-1 = \mu(2)$, which is the classical Moebius coefficient. The factor d demanded by the conjecture is what fails. Under R1 the true relation is $2B_2 = T_2 - T_1$, which has no integer form.

4 Scope

We refute clause (i) under the six readings R1–R3 and R1'–R3', for finite dynamical systems, at order 2. The witness g is a genuine permutation with both fixed points and a 2-cycle. We do not claim that every conceivable reading is refuted. In particular, a total count that weights each primitive k -orbit by n/k , namely $T_n = \sum_{k|n} (n/k)B_k$, would make the $\mu(d)d$ formula correct. That weighting does not appear in the text, so we do not treat it.

5 Lean formalization and axiom audit

The file `Conjecture5402/Basic.lean` (281 lines) imports only `Mathlib` (v4.33.1) and defines everything above in the namespace `C5402`. The main theorem is

```
theorem conjecture_5402_false (N : Nat) (hN : 2 <= N) :
  not (ConjecturedConversion perPts primOrbits N) /\
  not (ConjecturedConversion perPts exactPts N) /\
  not (ConjecturedConversion divOrbits primOrbits N) /\
  not (ConjecturedConversion cumPts primOrbits N) /\
  not (ConjecturedConversion cumPts exactPts N) /\
  not (ConjecturedConversion cumOrbits primOrbits N)
```

Two further theorems are `R1_no_integer_conversion` and `no_placement_of_muD`. For each of the three, `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.

Sources

[W1] Wikipedia, “Möbius inversion formula”, https://en.wikipedia.org/wiki/M%C3%B6bius_inversion_formula. The article states the formula as: if $g(n) = \sum_{d|n} f(d)$ for every integer $n \geq 1$, then $f(n) = \sum_{d|n} \mu(d) g(n/d)$ for every integer $n \geq 1$, and adds: “In effect, the original $f(n)$ can be determined given $g(n)$ by using the inversion formula.”